Ella Iles

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Education

Harvard University 2026 - Present
Ph.D. Astronomy at the Center for Astrophysics | Harvard & Smithsonian.

McGill University 2021-2025
B.Sc. Honours Physics,
Honours Thesis: Dark Photon and Axion Production in Magnetized White Dwarfs, Supervised by Prof. Katelin Schutz.
First Class Honours, Dean's Honour List.

Berlin British School 2019-2021
International Baccalaureate Diploma,
Score: 44/45.

Awards and Scholarships

Ashburn Fellowship, Harvard University 2026
Fellows comprise a cohort of 5 students a year who produce scholarship in, and creative solutions for, some of today's most fascinating questions across the sciences and the humanities. USD $32,000.

P.R. Wallace Prize in Physics, McGill University 2025
Awarded by the Faculty of Science Scholarships Committee on recommendation of the Department of Physics to a graduating student who has shown marked aptitude for studies in theoretical physics. CA $525.

Dean's Honour List, McGill University 2022–2025
CGPA in the top 10% of the Faculty of Science.

James F. Mathison Scholarship, McGill University 2024–2025
Scholarship awarded on the basis of distinguished academic standing to a student enrolled in the Honours Physics program. CA $3,100.

Best Presentation Award, Trottier Space Institute 2024
Top prize awarded at the Trottier Space Institute Summer Undergraduate Research Showcase. CA $125.

Undergraduate Summer Research Award (USRA), NSERC 2024
Funded by the Natural Sciences and Engineering Research Council of Canada (NSERC), CA $8,700.

Undergraduate Research Supplement, FRQ 2024
Funded by the Fonds de recherche du Québec (FRQ). CA $1,500.

Tomlinson Engagement Award for Mentoring, McGill University 2023, 2025
Mentoring Award given on the basis of excellence in teaching, learning and mentorship. CA 350×2.

Garnet and Jane Shaver Veteran's Scholarship, McGill University 2023–2024
Awarded to an outstanding undergraduate student on the basis of high academic standing by the Faculty of Science Scholarships Committee. CA $3,250.

J.W. McConnell Scholarship, McGill University 2021–2025
Renewable scholarship awarded based on exceptional academic and extracurricular achievements. CA $20,000.

Publications

D. Sarkar, **E. Iles**, A. Liu, "Clues from $\mathcal{Q}$: A Null Test for Line Intensity Mapping Cross-Correlations," submitted to Phys. Rev. D. arXiv:2512.09984.

N. Brahma, **E. Iles**, H. Scherer, K. Schutz, "Resonant Axion and Dark Photon Production in Magnetic White Dwarfs," 2026, Phys. Rev. D. 113, 083010.

E. Iles, S. Heeba, K. Schutz, "Dark Matter Direct Detection Experiments Are Sensitive to the Millicharged Background," 2025, Phys. Rev. Lett. 134, 121002.

Research Experience

Australian National University 2025
Collaboration with Prof. Stuart Wyithe's group at the Australian National University and Prof. Adrian Liu's Cosmic Dawn group developing a public Python package to supersize simulations of the 21 cm field using effective field theory.

McGill University Dark Matter Group 2024–2025
Honours Thesis Supervisor: Prof. Katelin Schutz.
Investigated detection potential of millicharged particle backgrounds; published in Phys. Rev. Lett. 134, 121002 (2025) arXiv:2407.21096.
Calculated dark photon and axion production in magnetized white dwarfs; published in Phys. Rev. D. 113, 083010arXiv:2509.07085.

McGill University Cosmic Dawn Group 2023–2025
Supervisor: Adrian Liu.
Investigated the noise properties of novel statistical estimators combining cosmological surveys conducted from four different spectral lines; submitted to Phys. Rev. D. arXiv:2512.09984

Advanced Laser Light Source Laboratory Feb–Mar 2023
Institut National de la Recherche Scientifique (INRS).
Aligned a 200 TW laser and contributed to building a new beamline extension to achieve world-leading intensities.

International Physics Tournament 2022–2023
Team Canada Member for McGill University.
Investigated plasma creation from graphite in a microwave, the "infinite lamp," and the "singing ice" phenomenon.

Teaching and Mentoring

Undergraduate Teaching Assistant, McGill University 2023, 2025
PHYS 102 (Introduction to Electromagnetism) and PHYS 351 (Honours Classical Mechanics II). Held weekly tutorials and office hours.

Private Tutor 2021–Present
Tutored International Baccalaureate and Advanced Placement students in mathematics, physics and chemistry; supported McGill students in Math 133 (Linear Algebra).

Leadership

Undergraduate Equity, Diversity and Inclusion Coordinator 2024–2025
McGill Physics Department. Organized EDI initiatives and resource accessibility projects with the McGill Society of Physics Students (MSPS).

Wilderness Canoe Trip Guide, Camp Wanapitei and Nahanni River Adventures 2022–2026
Guiding youth and adults on canoe trips in Northern Ontario, Quebec, Yukon and the Northwest Territories. Teaching outdoors skills, environmental stewardship and indigenous history.

Whitewater Executive, McGill Outdoors Club 2023–2025
Led kayaking trips and safety sessions on the Rouge and St. Lawrence Rivers; ran weekly "Learn to Roll" sessions at McGill pool.

Head Girl, Berlin British School 2021
Recognized for academic excellence and leadership.

Talks

Friday Lunch Talks, Australian National University October 2025
Introduction to using stars to look for beyond the standard model particles with light boson production in white dwarfs as a case study.

End of Summer Research Showcase, Trottier Space Institute August 2024
On millicharged particle production in the early universe and the possibility of detecting them using dark matter detection experiments.

Other Work Experience

Wilderness Canoe Trip Guide, Nahanni River Adventures July 2026–August 2026
Guiding canoe trips on the remote Nahanni River in the Northwest Territories. Responsible for safety, logistics, and group dynamics in a remote setting. Selected based on technical skills, paddling ability and leadership.

Whitewater Canoe Instructor, Madawaska Kanu Center May 2026–July 2026
Instructing beginner and intermediate whitewater canoe courses on the Madawaska River. Responsible for safety, lesson plan and progression of students.

Wilderness Canoe Trip Guide, Camp Wanapitei June 2022–2025
Guided youth on 6–30 day wilderness canoe expeditions in Northern Ontario and Quebec. Responsible for safety, logistics, route planning, and group dynamics. Selected for strong leadership, collaboration, and risk-management skills.

Editor for The Abstract, McGill Science Undergraduate Research Journal 2022–2023
Editor for the popular science blog The Abstract run by The McGill Undergraduate Research Journal.

Reforestation Worker, Folklore Contracting / Blue Collar Silviculture 2022–2023
Worked in British Columbia (Prince George, Fort St. James, Quesnel) planting up to 3,500 trees per day in remote and physically demanding conditions over two summers.

Antique Jewelry Sales 2017
Sold pieces of antique jewelry at flea markets in Ottawa. The collection had an emphasis on brutalist design.

Technical Skills

Programming: Python (NumPy, SciPy, Matplotlib), Mathematica, Java, LaTeX.
Research: Numerical simulation, data analysis, model development.
Languages: English (native), French (B2).

Certifications

National Lifeguard (NLS)
Certified lifeguard qualified in pool and open water rescue, first aid, and emergency response.

Wilderness First Responder (WFR)
Certified to handle medical emergencies in remote environments. Highest level of wilderness first aid without EMT training.

Whitewater Rescue Technician (WRT I–III)
Certified to perform whitewater rescues and guide whitewater canoe and kayak trips safely. Highest level of whitewater rescue certification.

Interests

Canoe Tripping: Longest trip was a 54-day self-sustained expedition from the Yukon into the Northwest Territories.
Whitewater Kayaking: Paddling on the Ottawa and Gatineau Rivers.
Reading Literature